

CEO tenure and firm performance: the effects of long-term tenure in major Japanese electrical equipment firms

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Abstract

Purpose – This study aims to examine the relationship between firm performance and CEO tenure in major Japanese electrical equipment firms, which operate in an industry that is important within the Asian corporate governance context.

Design/methodology/approach – Using a 26-year panel dataset (FY1999–FY2024) covering 621 firm-year observations from 24 major Japanese electrical equipment firms, this study estimates the tenure–performance relationship with fixed-effects regressions and cluster-robust standard errors, as indicated by the Hausman test.

Findings – The results reveal a horizontal S-shaped relationship between firm performance and CEO tenure. This pattern appears to be driven by a small number of founder-led firms with exceptionally long CEO tenures, and should be interpreted with caution given the limited observations and the focus on accounting-based performance.

Research limitations/implications – The findings extend Hambrick and Fukutomi's (1991) CEO life-cycle perspective by suggesting that, under certain conditions, long-tenured CEOs may again contribute positively to firm performance. The results may also be informative for Asian firms operating in environments where external governance mechanisms are relatively weak and CEO turnover is less market-driven.

Originality/value – First, by using long-term panel data from major Japanese firms, this study provides rare empirical evidence on the effects of CEO tenure on firm performance. Second, it identifies a polarized distribution of CEO tenure in Japanese electrical equipment firms, where extremely long tenures, concentrated in a small number of founder-led firms, coexist with short tenures often associated with post-appointment performance declines. These findings contribute to a deeper understanding of CEO succession practices in large Japanese corporations.

Keywords CEO tenure, Firm performance, Corporate governance, Founder CEO, Japanese electrical equipment firms, Nonlinear relationship, Asia

Paper type Research paper

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1. Introduction

The purpose of this study is to quantitatively examine the relationship between firm performance and CEO tenure in major Japanese electrical equipment firms operating in a key industry within the Asian corporate governance context. Numerous factors that affect firm performance have been identified, including strategic alliances (e.g. [Rothaermel et al., 2006](#)), outsourcing (e.g. [Kotabe et al., 2012](#)), corporate governance (e.g. [Bhagat and Bolton, 2019](#); [Shakri et al., 2025](#)) and corporate social responsibility (e.g. [Ho et al., 2024](#); [Yang et al., 2019](#)). These factors play an important role in shaping firm performance, and CEO leadership is considered highly influential in directing such initiatives, as CEOs can affect the entire organization through strategy formulation, implementation and leadership ([Hambrick and Quigley, 2014](#)). Prior research shows that CEOs systematically influence firm policies and organizational practices through their individual managerial styles

JEL classification – G34, M10, M12

Received 4 January 2026
Revised 3 April 2026
2 May 2026
Accepted 6 May 2026

(Bertrand and Schoar, 2003), and that their abilities and execution skills are strongly associated with firm performance (Kaplan *et al.*, 2012).

However, CEOs do not necessarily contribute to firm performance in the same way throughout their tenure. Because CEOs' knowledge, motivation and role can change over time, the length of their tenure may influence firm performance (Hambrick and Fukutomi, 1991). Focusing on this point, this study investigates the relationship between CEO tenure and firm performance.

Although several studies have examined the relationship between CEO tenure and firm performance, relatively few have focused on Japanese firms. For example, most Japan-based studies have examined CEO-related issues such as turnover (e.g. Abe, 1997; Kang and Shivdasani, 1995), compensation (e.g. Kaplan, 1994; Kato and Kubo, 2006) or corporate governance (e.g. Basu *et al.*, 2007; Sueyoshi *et al.*, 2010), rather than examining the dynamic tenure–performance relationship itself. This study narrows this gap by analyzing major Japanese electrical equipment firms – large, globally active companies whose long-term leadership structures provide a meaningful setting for examining the performance implications of CEO tenure in Asia.

The Japanese electrical equipment sector provides a particularly relevant setting for examining CEO tenure–performance dynamics because it combines continuous innovation cycles and strong global competition. These characteristics often require sustained strategic commitment and long-term resource allocation, which may amplify both the risks of late-tenure rigidity and the potential for strategic renewal under long-tenured leadership.

This study adopts a deliberately concise empirical design to highlight the core pattern of the tenure–performance relationship. The remainder of this paper is organized as follows: Section 2 reviews the prior literature and develops the hypotheses. Section 3 describes the analytical methods, followed by the presentation of the empirical results in Section 4. Section 5 provides a discussion of the findings, and Section 6 concludes the paper.

2. Literature review and hypothesis development

CEO tenure is a key observable CEO attribute that has attracted attention in the fields of management, accounting and finance (Darouichi *et al.*, 2021) and has been widely examined in empirical studies (e.g. Allgood and Farrell, 2000; Brochet *et al.*, 2021; Simsek, 2007). Many studies investigating its relationship with firm performance report an inverted U-shaped pattern when firm performance is plotted on the vertical axis and CEO tenure on the horizontal axis (e.g. Brochet *et al.*, 2021; Chikunda *et al.*, 2025; Henderson *et al.*, 2006; Miller and Shamsie, 2001).

Hambrick and Fukutomi (1991) offer a theoretical explanation for the mechanisms underlying this inverted U-shape through their CEO life-cycle model, which posits that CEOs progress through five stages over the course of their tenure:

1. response to mandate;
2. experimentation;
3. selection of an enduring theme;
4. convergence; and
5. dysfunction.

According to the authors, CEO activities are vigorous in the early stages as they seek legitimacy and acquire knowledge from internal and external stakeholders.

During the mid-tenure stage, performance tends to improve as CEOs concentrate on implementing their selected strategic themes. However, as CEO tenure lengthens, cognitive

rigidity and strategic inertia may emerge, increasing the likelihood of performance decline. The CEO life-cycle model proposed by [Hambrick and Fukutomi \(1991\)](#) extends the Upper Echelons perspective ([Hambrick and Mason, 1984](#)), which emphasizes managerial background characteristics in shaping strategic choices and firm outcomes.

[Brochet et al. \(2021\)](#) empirically validate the inverted U-shaped pattern and note that “[Hambrick and Fukutomi \(1991\)](#) is the only explicit theory of firm performance during CEO tenure” (p.50), reaffirming the model’s theoretical importance.

Studies focusing on Asian countries report mixed evidence regarding the relationship between CEO tenure and firm performance. While some studies find a statistically positive relationship, such as in Iran ([Ruhani and Imeni, 2025](#)) and Indonesia ([Triyani et al., 2020](#)), others document a negative relationship, as observed in China ([Liu and Jiang, 2020](#)). One possible explanation for these inconsistent findings is the variation in empirical approaches adopted in prior research, including differences in model specifications and sample characteristics ([Darouichi et al., 2021](#)). Beyond studies limited to Asia, recent research analyzing publicly listed firms in both emerging and developed economies – including CEOs with tenures of up to 48 years – suggests that very long-tenured CEOs, particularly those serving for more than 30 or even 40 years, may contribute positively to firm performance ([Nwafor et al., 2025](#)).

In addition, a recent study indicates that some long-tenured CEOs may overcome periods of stagnation and reconstruct their strategies in response to environmental changes ([Citrin et al., 2019](#)). According to [Citrin et al. \(2019\)](#), “The executives who appear on Harvard Business Review’s 2019 list of best-performing CEOs in the world show remarkable longevity in their roles; The average honoree has been CEO for 15 years, more than double the S&P 500 average in 2017 of 7.2 years” (p. 56). This suggests that long tenure does not necessarily result in deteriorating performance.

Prior research suggests that founder or family CEOs are often associated with superior firm performance, particularly when ownership and control are aligned ([Anderson and Reeb, 2003](#); [Villalonga and Amit, 2006](#)). In general, CEOs face performance-sensitive turnover, whereby poor performance increases the likelihood of dismissal ([Jenter and Lewellen, 2021](#)). However, in founder- or family-controlled firms, such disciplinary mechanisms may be weaker, allowing CEOs to remain in office longer and pursue long-term strategic objectives. At the same time, evidence indicates that CEO-specific abilities and execution skills play a critical role in shaping firm performance ([Kaplan et al., 2012](#)).

When such conditions hold – namely, insulation from short-term performance pressures combined with sufficient managerial capability – long-tenured CEOs may not simply drift into rigidity, but instead retain the discretion and capacity to undertake strategic reconfiguration at later stages of their tenure. Consequently, in contrast to the standard prediction of late-tenure decline driven by rigidity, certain long-tenured CEOs, particularly founders or family-affiliated leaders, may generate a recovery in firm performance at later stages of tenure.

Connecting these perspectives with [Hambrick and Fukutomi’s \(1991\)](#) model implies that, while many CEOs experience cognitive rigidity and strategic inertia in later stages, some long-tenured CEOs may successfully reconfigure their strategic themes in response to major environmental changes. Consequently, after reaching a mid-tenure peak and subsequently declining, performance for some long-tenured CEOs may rise again.

This argument also implies that the predicted late-tenure recovery should be contingent on governance conditions. In particular, if such recovery is primarily driven by founder or family control and weaker performance-sensitive turnover, the recovery observed at later stages of CEO tenure should attenuate or disappear in settings where these conditions are absent.

Taken together, these arguments suggest that while the general tendency of CEO tenure may follow an inverted U-shaped pattern, certain governance conditions may allow for a recovery in firm performance at later stages of tenure.

Based on the above discussion, the following hypothesis is proposed:

- H1.* Firm performance increases from the early to mid-tenure stage, then declines, but rises again for long-tenured CEOs. That is, it follows a reversed horizontal S-shaped pattern (an inverted U followed by a U-shaped pattern; see [Figure 1](#)).

Rather than focusing on a strict global distinction between reversed horizontal S-shaped and horizontal S-shaped patterns, this study emphasizes the underlying tenure–performance dynamics, in which firm performance may evolve nonlinearly across different stages of CEO tenure, potentially varying across governance contexts.

3. Research methodology

3.1 Data and sample

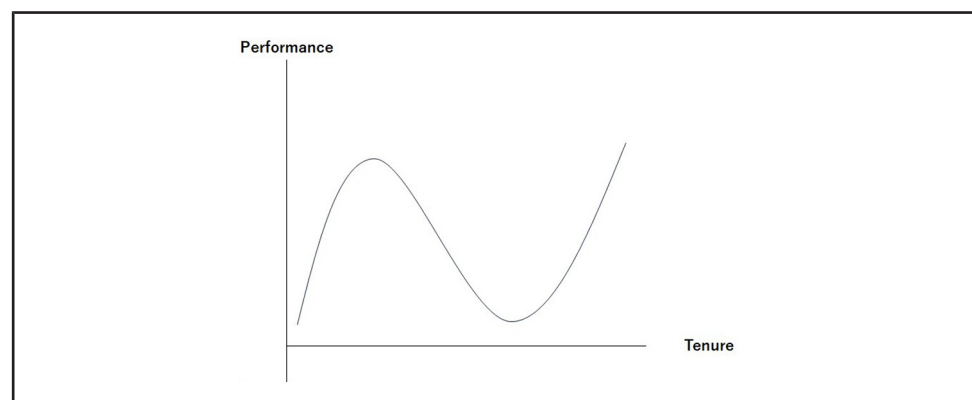
The sample of this study consists of 24 major Japanese firms classified under “Electric Appliances” in the Tokyo Stock Exchange industry classification. These firms are the top 24 out of the 25 largest firms by sales in FY2024 (for firms with the March fiscal year–end) and include Sony Group, Hitachi and Panasonic Holdings. Kioxia Holdings, which ranked 15th, was excluded from the analysis because it was listed on the Tokyo Stock Exchange Prime Market only in December 2024 and had publicly available data for FY2024 alone at the time of data collection.

The top 25 firms were selected because the largest sales gap – about 20% – within the 7th–160th rank range appears between the 25th and 26th firms, providing a natural cutoff that minimizes size imbalance. The top six global firms are indispensable and therefore included regardless of size differences. Focusing on the largest firms is also appropriate for the purpose of this study, as it allows for a relatively homogeneous comparison group in terms of global market exposure, organizational complexity and governance structures. Given that firm size varies substantially within the Japanese electrical equipment sector, including smaller firms would likely introduce significant heterogeneity that may obscure tenure–performance dynamics.

The study covers 26 years (FY1999–FY2024), for which consolidated financial statements are available. Owing to a later listing, one firm contributes only 23 years of data, yielding 621 firm-year observations. All sample firms have a March fiscal year-end.

Financial data were obtained from Nikkei NEEDS Financial QUEST on a consolidated basis, using financial statements disclosed in annual securities reports, with IFRS-based figures used for firms adopting IFRS. Missing information not available from this data base,

Figure 1 Reversed horizontal S-shaped relationship (an inverted U followed by a U-shaped pattern)



including financial data and variables related to CEOs and firms, was manually collected from annual securities reports. Because a CEO is responsible for the entire corporate group, including consolidated subsidiaries, the use of consolidated financial data is appropriate for evaluating firm performance.

It should also be noted that very long CEO tenures are highly concentrated in a small number of firms. As shown later in [Figure 3](#), observations for tenures 27 years or more are driven by a single firm. This concentration should be taken into account when interpreting the results, particularly in the long-tenure range.

3.2 Variable definition and estimation method

3.2.1 Dependent variable. The dependent variable is ROA (net income ÷ total assets), a widely used accounting-based performance indicator (e.g. [Henderson et al., 2006](#)). Recent studies examining the relationship between CEO tenure and firm outcomes also adopt ROA as a primary measure (e.g. [Chikunda et al., 2025](#); [Nwafor et al., 2025](#); [Ruhani and Imeni, 2025](#)). Although Tobin's Q is frequently used in related research (e.g. [Brochet et al., 2021](#)), it reflects market valuation, whereas ROA directly captures internal profitability. Because this study focuses on how CEO decision-making and organizational governance influence internal firm performance rather than investor sentiment or market conditions, ROA was selected as the principal performance indicator.

3.2.2 Independent variable. The main independent variable is CEO tenure (CTN), measured as the completed number of years in office as of fiscal year-end (March 31). If a CEO assumes office mid-year (e.g. June), the fiscal year-end (the following March) is counted as part of the predecessor CEO's tenure.

In many Japanese firms, the president holds ultimate responsibility for daily executive management even when a chairman position exists. Therefore, unless the annual securities report explicitly designates a "CEO," the president is treated as the CEO. CEO tenure and age were collected from annual securities reports. Across the sample period, 112 individuals served as CEO, of whom 92 had clearly identifiable tenure. The average confirmed tenure was 7.2 years.

To test potential nonlinear relationships between CEO tenure and performance, squared and cubed terms of tenure were included. Guided by the hypothesis that performance shifts from an inverted U-shape to a U-shape over the course of a CEO's life cycle, a cubic specification was employed.

3.2.3 Control variables. The following control variables were employed:

- CEO Power Centralization (PWR): Coded 1 if the CEO is listed first in the annual securities report or concurrently serves as board chair/presiding officer; otherwise 0, following [Fizel and Louie \(1990\)](#). [Quigley and Hambrick \(2012\)](#) show that firm performance (ROA) changes significantly when the former CEO remains as chair of the board. This measure captures the notion of structural power emphasized in recent studies (e.g. [Maharani et al., 2025](#)).
- CEO Age (CAG): Widely used in studies examining CEO tenure (e.g. [Brochet et al., 2021](#); [Henderson et al., 2006](#); [Simsek, 2007](#)), measured as age at fiscal year-end.
- Firm Age (FAG): A basic firm characteristic (e.g. [Brochet et al., 2021](#); [Henderson et al., 2006](#)), measured as the number of years since establishment at fiscal year-end.
- Firm Size (SIZE): The natural logarithm of the number of employees (e.g. [Henderson et al., 2006](#); [Li and Patel, 2019](#)).

- Sales per Employee (SPE): A standard measure of labor productivity (e.g. [Kotabe et al., 2012](#); [Mol, 2005](#)). Because the variable is skewed, the natural logarithm of sales (in million yen) divided by the number of employees was used.
- Leverage (LEV): A commonly used indicator of financial leverage, calculated as total debt divided by total assets (e.g. [Chikunda et al., 2025](#); [Khan et al., 2020](#)).

Given that the sample firms are globally active manufacturing companies in a high-technology industry, two additional control variables were included:

1. Overseas Sales Ratio (OSSR): Overseas and export sales divided by total sales.
2. R&D Intensity (RDI): R&D expenditure divided by sales ([Kotabe et al., 2012](#)). Because the variable is highly skewed, its natural logarithm was used.

Year dummies were also included to control for macroeconomic and industry-wide fluctuations.

3.2.4 Estimation method. The analyses were conducted using Stata 14. Based on the results of the Hausman test, the fixed-effects model was selected. Because CEO tenure varies widely across firms, their error structures also differ, which can cause conventional standard errors to overstate significance. Therefore, cluster-robust standard errors were used.

Using the above variables, the fixed-effects regression model for firm i ($i = 1, \dots, 24$) in year t ($t = 1999, \dots, 2024$), with CTN as the main explanatory variable, is specified as [equation \(1\)](#):

$$ROA_{i,t} = \beta_0 + \beta_1 CTN_{i,t} + \beta_2 CTN2_{i,t} + \beta_3 CTN3_{i,t} + \beta_4 PWR_{i,t} + \beta_5 CAG_{i,t} + \beta_6 FAG_{i,t} + \beta_7 SIZE_{i,t} + \beta_8 SPE_{i,t} + \beta_9 LEV_{i,t} + \beta_{10} OSSR_{i,t} + \beta_{11} RDI_{i,t} + YearFE + F_i + u_{i,t} \quad (1)$$

While the cubic specification allows for flexible nonlinear patterns, it should be noted that, unlike quadratic specifications, where established procedures exist to evaluate curvature and turning points (e.g. [Haans et al., 2016](#); [Lind and Mehlum, 2010](#)), there is no widely accepted method for rigorously testing the validity of a cubic functional form in this setting.

4. Results

[Table 1](#) reports the descriptive statistics and correlation matrix for the variables used in this study. Although the correlation coefficient between leverage (LEV) and firm size (SIZE) is relatively high at 0.592, it remains below the commonly cited threshold for multicollinearity ($|r| < 0.7$). Thus, it is unlikely to pose any serious multicollinearity concerns for the regression estimates. CEO tenure (CTN) ranges from 1 to 48 years, with a mean of 6.2 years. CEO age (CAG) ranges from 45 to 80, with a mean of 61.7 years.

[Table 2](#) presents the estimation results for four regression models using ROA as the dependent variable. Model 1 serves as the baseline specification. Building on this, Model 2 adds CTN, Model 3 adds the squared term of CTN (CTN2), and Model 4 additionally incorporates the cubic term (CTN3). Model 4 is the main model for testing the hypothesis of this study.

CTN becomes statistically significant only in Model 4, where the linear term is negative, the squared term is positive, and the cubic term is negative; all are significant at the 5% level. Examining AIC and BIC, Models 1–3 show little difference in their values. By contrast, Model 4 exhibits AIC and BIC values that are lower by approximately 13–15 points, indicating clearly improved model fit. This demonstrates that including the cubic term substantially enhances explanatory power. Therefore, Model 4, which incorporates the polynomial terms up to the third order, is judged to provide the best overall fit.

Regarding the control variables, firm age (FAG) is negative and significant at the 10% level; sales per employee (SPE) is positive and significant at the 1% level; and leverage (LEV) and R&D intensity (RDI) are negative and significant at the 1% and 5% levels, respectively. The

Table 1 Descriptive statistics and correlation matrix (*n* = 621)

Variable	Mean	SD	Min.	Max.	ROA	CTN	PWR	CAG	FAG	SIZE	SPE	LEV	OSSR	RDI
ROA	0.038	0.054	-0.371	0.231	1									
CTN	6.232	7.399	1	48	0.096	1								
PWR	0.430	0.495	0	1	0.090	0.400	1							
CAG	61.747	5.481	45	80	-0.163	0.271	0.209	1						
FAG	72.375	23.498	26	125	-0.208	-0.308	-0.100	0.259	1					
SIZE	10.716	1.202	7.382	12.860	-0.346	-0.031	-0.083	0.375	0.491	1				
SPE	3.316	0.618	1.538	4.855	0.263	-0.307	-0.137	-0.173	-0.009	-0.380	1			
LEV	0.484	0.209	0.038	1.020	-0.535	-0.095	0.077	0.346	0.414	0.592	-0.166	1		
OSSR	0.600	0.209	0.13	0.98	0.205	0.196	0.312	-0.031	-0.327	-0.146	-0.161	-0.296	1	
RDI	-2.923	0.412	-4.143	-1.090	-0.123	-0.078	-0.034	-0.130	-0.132	-0.256	0.165	-0.140	0.321	1

Table 2 Fixed-effects regression results with firm-clustered standard errors (dependent variable: ROA)

Model	1	2	3	4
CTN		0.000399 [0.79]	0.000669 [0.55]	−0.003195 [−2.07]**
CTN2			−0.000009 [−0.38]	0.000296 [2.12]**
CTN3				−0.000005 [−2.07]**
PWR	0.018472 [1.92]*	0.01719 [1.73]*	0.01681 [1.68]	0.015851 [1.60]
CAG	−0.001193 [−2.01]*	−0.001348 [−1.86]*	−0.001367 [−1.80]*	−0.00101 [−1.60]
FAG	−0.001761 [−1.91]*	−0.001751 [−1.88]*	−0.001765 [−1.90]*	−0.001849 [−1.94]*
SIZE	0.018701 [0.98]	0.018151 [0.96]	0.018153 [0.95]	0.020031 [1.07]
SPE	0.069156 [2.57]**	0.069654 [2.60]**	0.069982 [2.62]**	0.075776 [2.88]**
LEV	−0.137074 [−2.98]**	−0.136369 [−2.94]**	−0.136125 [−2.95]**	−0.139041 [−2.92]**
OSSR	0.091812 [1.49]	0.097467 [1.57]	0.099427 [1.58]	0.098553 [1.56]
RDI	−0.051326 [−2.64]**	−0.05182 [−2.64]**	−0.051539 [−2.66]**	−0.052139 [−2.79]**
_cons	−0.347391 [−1.54]	−0.341354 [−1.53]	−0.341627 [−1.52]	−0.389056 [−1.71]
Year FE	Yes	Yes	Yes	Yes
N	621	621	621	621
R ²	0.5008	0.5018	0.5021	0.5125
Adj R ²	0.4736	0.4738	0.4732	0.4834
AIC	−2471.22	−2472.58	−2472.86	−2486.04
BIC	−2369.30	−2370.66	−2370.94	−2384.12
Hausman (p-value)	0.0001	0.0005	0.0006	0.0000

Note(s): [*t*-values]. **p* < 0.1; ***p* < 0.05; ****p* < 0.01

variable representing efficiency shows a positive association, whereas the variables representing cost burdens show negative associations. These results are theoretically consistent. The coefficient for the overseas sales ratio (OSSR) is positive, as expected, but not statistically significant.

When incorporating squared or cubic terms into the regression equation, care must be taken in interpreting the estimated coefficients. Accordingly, to visually examine the shape of the relationship between firm performance and CEO tenure in the hypothesis-testing model (Model 4), a theoretical curve based on the regression estimates was plotted. Figure 2 shows this curve with ROA on the vertical axis and CTN on the horizontal axis. Figure 3 presents the frequency distribution of CTN.

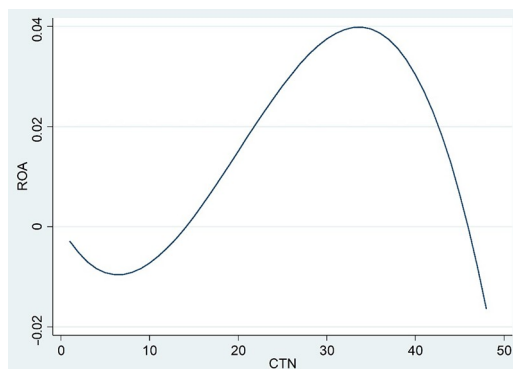
Figure 2 illustrates the curve generated by varying only CTN while holding constant the effects of control variables and year dummies. Thus, the ROA values in the figure do not replicate the observed values precisely. Nonetheless, the curve's shape reflects the estimates from Model 4. The regression equation yields a minimum and maximum ROA at 6.4 years and 33.6 years of tenure, respectively.

Importantly, both the shape of the curve and the estimated turning points (6.4 and 33.6 years) should be interpreted with caution, as the long-tenure segment is concentrated in a small number of firms – many of which are founder-led – and the analysis relies on an accounting-based performance measure (ROA). Given the limited number of observations in the long-tenure range, these estimated turning points are subject to sampling uncertainty. Accordingly, the cubic curve should be viewed as an illustrative approximation of tenure–performance dynamics rather than as a definitive functional form.

While additional descriptive visualizations (e.g. binned plots or kernel-based representations) could offer further intuition, their usefulness in this setting is limited due to the sparse distribution of long-tenure observations – particularly in the long-tenure tail – and their sensitivity to arbitrary grouping choices. Accordingly, the analysis prioritizes regression-based inference, with the fitted curve serving as a simplified visual summary of the estimated relationship.

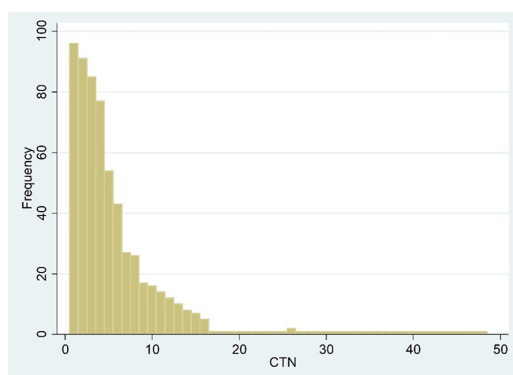
The hypothesis derived from prior studies predicts that CEO tenure raises firm performance from the early to mid-stages, followed by a decline, and then a subsequent rise among

Figure 2 Theoretical ROA–CTN curve



Note(s): The curve is based on varying CEO tenure while holding the effects of control variables and year dummies constant. Observations in the long-tenure range are limited; therefore, the tail of the curve should be interpreted with caution

Figure 3 CTN frequency distribution



Note(s): The figure shows the frequency distribution of CEO tenure. Observations in the long-tenure range are limited

long-tenured CEOs – a pattern resembling a reversed horizontal S-shape. However, the curve implied by the estimates in [Figure 2](#) suggests the opposite pattern (a horizontal S-shaped relationship): performance decreases until 6.4 years, rises thereafter to a peak at 33.6 years, and then declines again. This phenomenon will be examined in the discussion section.

To further examine the role of extreme long-tenure observations, the analysis was reestimated after excluding the firm with the longest CEO tenure. The results indicate that the nonlinear pattern changes and becomes closer to the theoretically predicted reversed horizontal S-shape, although statistical significance is reduced. This suggests that the full-sample pattern may be disproportionately influenced by a single firm with unusually long CEO tenure, rather than reflecting a general pattern across firms, and supports the

argument that late-tenure performance dynamics are contingent on specific governance conditions. Detailed results are reported in [Table A1](#) in the [Appendix](#).

On the other hand, controlling for a founder CEO indicator, the nonlinear pattern remains qualitatively similar, and the estimated coefficients retain the same signs as in the main specification although statistical power is limited. This result is consistent with the argument that governance structures associated with founder leadership may shape tenure–performance dynamics (see [Table A2](#) in the [Appendix](#)). Additional analyses incorporating interaction terms between CEO tenure and founder status yield qualitatively similar implications, with the cubic term remaining negative and statistically significant, suggesting that tenure–performance dynamics may vary depending on governance context (see [Table A3](#) in the [Appendix](#)).

Additional piecewise estimations (results not reported for brevity) yield partially consistent patterns, while suggesting that the late-tenure segment is influenced by a limited number of observations, particularly at the extreme end of the tenure distribution. Consistent with this, further leave-one-firm-out analyses (not reported) confirm that the estimated nonlinear pattern is broadly stable across firms, with the exception of the firm with the longest CEO tenure, as discussed above. These findings reinforce the interpretation that the nonlinear relationship is contingent on specific governance contexts.

Overall, the robustness analyses indicate that the estimated cubic pattern – particularly in the long-tenure range – is sensitive to the inclusion of extreme-tenure observations, as excluding the firm with the longest CEO tenure shifts the curve toward a reversed horizontal S-shape. At the same time, a consistent pattern emerges across alternative specifications: firm performance varies across CEO tenure stages, indicating a broadly nonlinear relationship. While the precise shape of the curve is not stable, the presence of tenure-related performance dynamics persists. These findings suggest that the tenure–performance relationship is nonlinear but context-dependent, particularly with respect to governance conditions and the influence of extreme-tenure observations.

5. Discussion

Given the illustrative nature of the cubic curve and the limited number of long-tenure observations, the discussion focuses on interpreting the broad patterns implied by the estimates rather than the precise shape of the curve.

The results presented in Section 4 indicate that the cubic specification estimated in Model 4 is statistically significant, implying a complex relationship between CEO tenure and ROA. However, as shown in [Figure 3](#), the number of firms with long-tenured CEOs is limited, suggesting that the shape of the curve in the long-tenure range may be heavily influenced by a small number of long-tenured CEO cases. Thus, while the curve illustrates the tendency implied by the estimated model, it cannot be uniformly generalized to all firms. While the full-sample results suggest an initial decline in performance following CEO succession, the supplementary analysis indicates that this pattern is not uniform and may vary depending on firm-specific governance conditions. Accordingly, the following discussion uses the full-sample results as a baseline reference, while interpreting them in light of these contextual limitations.

Given this background, the scope of the discussion is necessarily limited. Nevertheless, based on the regression results, two characteristics of major Japanese electrical equipment firms can be highlighted.

First, long CEO tenures are predominantly observed among founders or CEOs from founding families. Although the estimated peak ROA occurs at 33.6 years of tenure in the cubic curve, data for tenures of 17 years or more come from only three of the 24 firms. In particular, observations for tenures of 27 years or more are derived from just one firm. Consequently, the long-tenure portion of the curve is likely to be strongly shaped by these

few firms. This reflects a distinctive feature of the Japanese electric appliance industry, in which founders or founding-family members often remain in the CEO role over long periods. That is also commonly observed among large Asian corporations characterized by concentrated ownership and strong founding-family influence. Prior studies show that CEO turnover tends to be significantly lower in family-controlled firms (Tsai *et al.*, 2006), a governance feature that remains prevalent in many Asian economies. In such contexts, CEOs, often founders or members of founding families, are more likely to remain in office for extended periods. This pattern is observed also in other major Asian electronics firms, such as Samsung and Foxconn.

Second, firm performance tends to decline shortly after a new CEO takes office. Recent studies using ROA as the dependent variable show that CEO tenure tends to be positively significant, due to the accumulation and application of firm- and industry-specific knowledge that enables stable strategic decision-making (Nwafor *et al.*, 2025; Ruhani and Imeni, 2025). However, Citrin *et al.* (2019) note that CEOs often experience the “sophomore slump.” The regression results suggest that, in some Japanese electrical equipment firms, recovering from this slump may take time, and stagnant performance may lead to frequent CEO turnover. For example, among the firms included in the 26-year data set, five firms experienced six or more CEO transitions (four firms with six transitions, one firm with seven). Their average ROA was 1.2%, which is 2.6 percentage points lower than the sample mean of 3.8%. This substantial difference implies that deteriorating performance may trigger CEO replacement (Chijoke-Mgbame *et al.*, 2023; Jenter and Lewellen, 2021; Miller, 1991). This offers a practical implication that, rather than suggesting that executive team support is uniformly important across all stages of CEO tenure, the findings imply that the role of internal managerial support may vary over time. In particular, the results suggest that support from the executive team is especially important in the early stages of a CEO’s tenure, when firm performance tends to stagnate following leadership transition.

To further explore potential causes, the average value of PWR, an indicator of power concentration, is informative. Among the five firms with frequent CEO turnover, the average PWR was 0.285, which is substantially lower than the overall average of 0.430. Although PWR was only weakly supported statistically in Models 1 and 2 (significant at the 10% level), and the evidence is insufficient to make a strong claim about power concentration, the findings suggest that concentrating authority in the CEO, typically as the chair of the board, may enhance firm performance in large Japanese electrical equipment firms. This may also apply to other Asian firms, but further study is required.

Taken together, these observations point to the importance of governance structures and managerial contexts in shaping tenure–performance dynamics, thereby providing a basis for connecting the empirical patterns to broader theoretical arguments.

The findings in this study should be interpreted as contingent on sectoral and ownership contexts rather than universally applicable. The observed late-tenure upturn is most likely to arise in settings characterized by concentrated ownership and relatively weak performance-sensitive turnover, where CEOs, particularly founders or family members, can retain discretion over extended periods (Anderson and Reeb, 2003; Villalonga and Amit, 2006). In contrast, in widely held firms with active boards and strong performance-sensitive monitoring, such late-tenure recovery is less likely to emerge, as underperforming CEOs are more likely to be replaced (Jenter and Lewellen, 2021).

Beyond governance-based explanations, alternative mechanisms may also contribute to the observed performance dynamics. For example, innovation cycles, long-horizon R&D investments or international strategic reconfigurations may generate improvements in firm performance at later stages of CEO tenure, even in the absence of founder-driven renewal. Consistent with this view, prior research shows that CEO ability and execution skills are important determinants of firm performance (Kaplan *et al.*, 2012), and that persistent

managerial styles can shape firm policies over time ([Bertrand and Schoar, 2003](#)). These factors suggest that late-tenure performance recovery may reflect not only governance structures but also heterogeneity in managerial capabilities and strategic contexts.

From a practical perspective, the findings suggest that governance responses may need to vary across stages of CEO tenure. Greater board support may be important in the early stages, while maintaining performance-sensitive monitoring in later stages may help distinguish strategic renewal from managerial entrenchment. Such stage-contingent approaches should be interpreted in light of firm-specific governance contexts.

6. Conclusion

The relationship between firm performance and CEO tenure is difficult to capture using simple linear or quadratic forms, and generalization across contexts remains challenging. Against this background, the present study examined whether a cubic specification can better describe the tenure–performance relationship.

This study makes two contributions to the existing literature. First, the findings extend [Hambrick and Fukutomi's \(1991\)](#) CEO life-cycle perspective by suggesting that, under certain conditions, long-tenured CEOs may again contribute positively to firm performance. Using long-term data from major Japanese firms in the electrical equipment industry, this study quantitatively assessed the impact of CEO tenure on firm performance. Longitudinal panel analyses focusing on Japanese firms remain relatively scarce in international journals, and this study helps address that gap.

Second, the study highlights a polarized distribution of CEO tenure in Japanese electrical equipment firms, where extremely long tenures coexist with relatively short tenures that are often associated with performance declines following CEO appointment. This finding contributes to a better understanding of CEO succession practices in large Japanese corporations.

Based on the discussion above, these findings may be informative for Asian firms operating in environments where external governance mechanisms are relatively weak and CEO turnover is less market-driven.

This study has two main limitations. First, because the analysis focuses on major Japanese electrical equipment firms, caution is required when generalizing the results to other industries or to small and medium-sized firms. The Japanese electrical equipment sector includes firms that differ in innovation cycles, capital intensity and global competition, such as component manufacturers, conglomerates and consumer electronics producers. Such differences may influence how CEO tenure relates to firm performance. Accordingly, the observed tenure-performance patterns may partly reflect sector-specific dynamics, and direct comparisons with other industries remain limited. Further research is needed to assess the generalizability of the findings.

Second, the study relies on firm-level financial data from securities reports and does not directly measure CEO decision-making processes or the actual degree of power concentration. In addition, CEO social networks may play an important role in shaping sustainable governance in Asian firms ([Cahyono et al., 2025](#)). To identify causal mechanisms, such as how decision-making speed and authority distribution affect firm performance, future research would require additional variables and complementary data, including detailed governance information or qualitative evidence.

Declaration of generative AI and AI-assisted technologies in the manuscript

During the preparation of this work the author used ChatGPT in order to improve readability and clarity of the English text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Appendix

Table A1 Re-Estimation excluding the firm with the longest CEO tenure

Model	3a	4a
CTN	−0.001486 [−1.47]	0.001582 [1.31]
CTN2	0.000074 [1.87]*	−0.000316 [−1.94]*
CTN3		0.000012 [2.33]**
PWR	0.015345 [1.54]	0.01535 [1.55]
CAG	−0.000664 [−1.00]	−0.00073 [−1.08]
FAG	−0.002019 [−2.08]**	−0.002047 [−2.10]**
SIZE	0.015175 [0.73]	0.015646 [0.76]
SPE	0.086185 [3.02]***	0.089027 [3.08]***
LEV	−0.137465 [−2.79]**	−0.137552 [−2.82]***
OSSR	0.10187 [1.64]	0.09964 [1.63]
RDI	−0.046491 [−2.60]**	−0.044584 [−2.45]**
_cons	−0.365545 [−1.50]	−0.372069 [−1.54]
Year FE	Yes	Yes
N	595	595
R ²	0.5252	0.5275
Adj R ²	0.4963	0.4979
AIC	−2377.10	−2379.98
BIC	−2280.55	−2283.43
Hausman (<i>p</i> -value)	0.0000	0.0000

Note(s): [*t*-values]. **p* < 0.1; ***p* < 0.05; ****p* < 0.01. The coefficients on the cubic terms (CTN, CTN2, CTN3) exhibit a positive–negative–positive sign pattern, consistent with a reversed horizontal S-shape

Table A2 Re-Estimation including a founder CEO indicator

Model	3b	4b
CTN	0.000605 [0.52]	−0.002833 [−1.82]*
CTN2	−0.000016 [−0.68]	0.000262 [1.89]*
CTN3		−0.000004 [−1.89]*
FND	0.048084 [1.89]*	0.02222 [1.00]
PWR	0.015984 [1.60]	0.015567 [1.57]
CAG	−0.001384 [−1.76]*	−0.001054 [−1.62]
FAG	−0.001768 [−1.89]*	−0.001842 [−1.93]*
SIZE	0.02057 [1.07]	0.020958 [1.09]
SPE	0.072454 [2.72]**	0.076331 [2.89]***
LEV	−0.136417 [−2.99]***	−0.138881 [−2.94]***
OSSR	0.097243 [1.55]	0.097632 [1.55]
RDI	−0.050911 [−2.64]**	−0.051788 [−2.78]**
_cons	−0.373059 [−1.63]	−0.398772 [−1.71]
Year FE	Yes	Yes
N	621	621
R ²	0.5059	0.5132
Adj R ²	0.4763	0.4832
AIC	−2477.66	−2486.94
BIC	−2375.74	−2385.02
Hausman (<i>p</i> -value)	0.0007	0.0001

Note(s): [*t*-values]. **p* < 0.1; ***p* < 0.05; ****p* < 0.01. FND (Founder CEO dummy) is an indicator variable equal to 1 if the CEO is the founder of the firm, and 0 otherwise. Founder CEOs are identified for two firms, yielding 27 firm-year observations

Table A3 Re-Estimation including founder CEO indicator and interaction terms

Model	3c	4c
CTN	0.001268 [1.15]	−0.002072 [−1.57]
CTN2	−0.000097 [−1.58]	0.000174 [1.68]
CTN3		−0.000004 [−2.29]**
FND	−0.017185 [−0.43]	−0.039662 [−1.04]
CTN × FND	0.004098 [1.63]	0.003943 [2.36]**
PWR	0.016403 [1.63]	0.015985 [1.59]
CAG	−0.000998 [−1.42]	−0.000694 [−1.05]
FAG	−0.001752 [−1.90]*	−0.001824 [−1.93]*
SIZE	0.020979 [1.10]	0.021338 [1.12]
SPE	0.078325 [2.93]***	0.081842 [3.02]***
LEV	−0.135028 [−2.97]***	−0.137456 [−2.92]***
OSSR	0.090669 [1.49]	0.091293 [1.50]
RDI	−0.047739 [−2.61]**	−0.048705 [−2.74]**
_cons	−0.408885 [−1.75]*	−0.432327 [−1.83]*
Year FE	Yes	Yes
N	621	621
R ²	0.5128	0.5197
Adj R ²	0.4828	0.4892
AIC	−2486.45	−2495.19
BIC	−2384.53	−2393.27
Hausman (p-value)	0.0008	0.0000

Note(s): [t-values]. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. An interaction term between CEO tenure (CTN) and founder indicator (FND) is included

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